USER GUIDE · IOS · IPADOS · MACOS

RadiX Dose v2.5

A reference absorbed-dose calculator for external beam radiotherapy.

- **Reference Dosimetry** — release-ready, verified against several known cases (use with caution)
- **TPS Commissioning** — in development, preview only

RadiX Dose turns a chamber reading into a verified absorbed dose to water — and keeps every calculation, with the equipment and calibrations behind it, in one auditable place. It is built for the medical physicist doing reference dosimetry and periodic output checks at the linac.

It runs natively on iOS (iPhone and iPad) and macOS, works fully offline, and stores everything on-device. Under the hood it is one shared equipment library — machines, electrometers, detectors and beams (photon and electron) — and a single dose engine that computes the reference absorbed dose to water, presented as **two features**, of which only the first is clinically ready.

01 Two features, one engine

You pick a feature from the home menu (*Choose a Feature*). Both draw on the same facilities, machines, detectors and protocols — all defined by you in the **Settings** menu — but they sit at very different stages of maturity, and the app colours them accordingly: **blue** for the released tool, **gold** for the one still in development.

RELEASE-READY

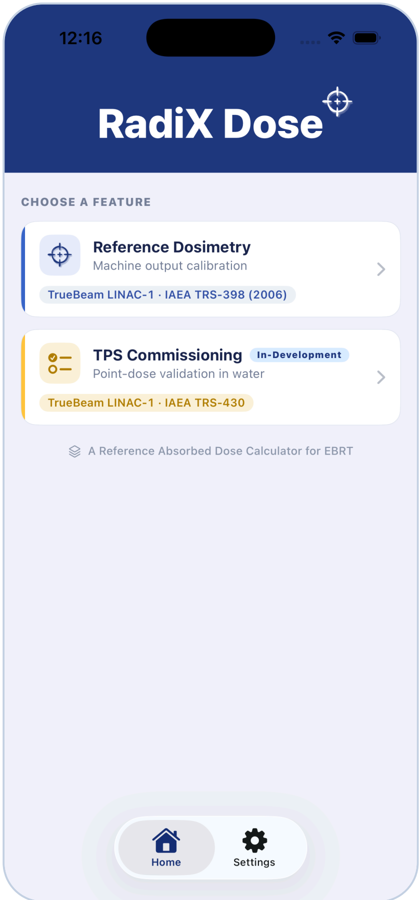
Reference Dosimetry

The core of the app. Compute absorbed dose to water from a chamber reading under IAEA TRS-398 or AAPM TG-51, check output against expected, and save an auditable record. Fully implemented and verified against several known reference cases — always confirm with your own independent check.

IN-DEVELOPMENT

TPS Commissioning

Point-dose validation of a treatment planning system against measurement, per IAEA TRS-430 — using the reference dose from IAEA TRS-398 or AAPM TG-51. **A working preview, not a validated clinical tool** — the workflow and acceptance logic are still settling. Do not rely on it for release decisions.



The home screen — pick Reference Dosimetry (released) or TPS Commissioning (preview).

02 The setup model — build your clinic once

Everything a calculation needs is described once, as a nested library under **Settings**. A result is only as trustworthy as the equipment record behind it, so this hierarchy is the app's backbone. One facility and one machine are *active* at a time; the whole app scopes to them.

SETUP HIERARCHY — SETTINGS

FACILITY **Hospital · department**

MACHINE **Linac · code of practice · setup & dose-reporting point**

Detector library

- Chamber from a built-in catalog (41 models, 6 manufacturers) or user-defined
- Manufacturer, serial number, cavity volume & length
- $N_{D,w}$ certificate + date + validity
- ^{60}Co *or* cross-calibration route — offered for every chamber
- Correction factors stored per chamber & energy

Electrometer library

- Manufacturer, model, serial number
- Optional calibration factor k_{elec} (default 1.000)
- Pairs with the chamber's $N_{D,w}$ certificate

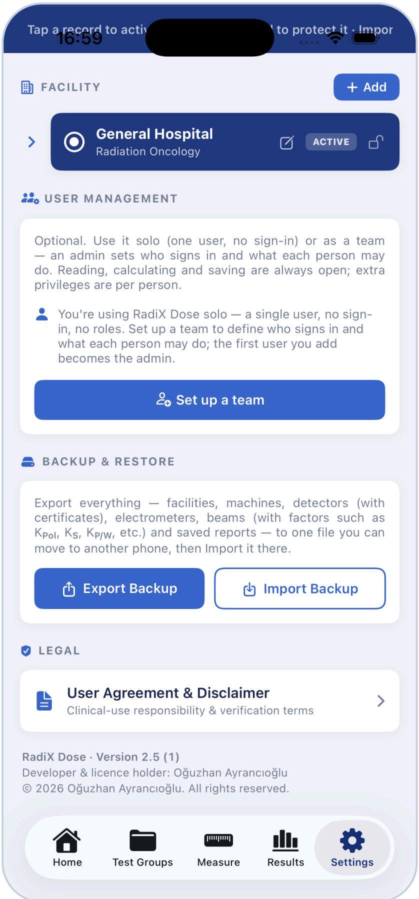
Beam library

- Photon & electron energies (incl. FFF)
- Beam-quality index per protocol ($TPR_{20,10}$ · $\%dd(10)_x$ · R_{50})
- **Calibration conditions:** field size, delivered MU & expected dose
- PDD / TMR at the reference depth (for d_{max} normalisation)

Each machine carries its own **code of practice**, **SSD/SAD setup**, and **dose-reporting point** (see section 03). Photon-only machines (Halcyon, Unity) and special-field machines (CyberKnife, TomoTherapy) are recognised and route their dosimetry accordingly — though the special-field machines are not yet fully validated, so use them with care; your feedback is welcome and will be applied where needed. Every delete — facility, machine, detector, beam — is confirmed before it cascades.

Calibration expiry, watched for you

Each certificate carries a date and a validity period. Detector cards turn **yellow at 3 months**, **red at 1 month**, and read **EXPIRED** past due — a warning only, never a block. You decide what to calculate with.



Settings — your facility, machines, chambers, and beam libraries in one place.

03 The calculation pipeline

This is the engine. A raw chamber reading is corrected for every influence quantity, scaled by the chamber's calibration and beam-quality factor, and turned into absorbed dose to water — then compared to what the beam *should* deliver.

FROM READING TO DOSE

INPUT

Reading + conditions

Up to 3 charge readings (their mean, `M`), temperature & pressure, chamber & beam.

↓

CORRECT

Influence factors

K_{TP} K_{Pol} K_S K_{Elec} $K_{P/W}$ K_{Vol}

↓

SCALE

Calibration & quality

$N_{D,w}$ K_Q

Chamber's dose-to-water factor and its beam-quality correction.

↓

RESULT

Absorbed dose $D_{w,Q}$

Dose to water, reported at the machine's chosen point (section below).

↓

CHECK

Output vs expected

Measured cGy/MU against the beam's calibration condition → % deviation.

corrected reading

$$M = M_{raw} \cdot K_{TP} \cdot K_{Pol} \cdot K_S \cdot K_{Elec}$$

absorbed dose to water

$$D_{w,Q} = M \cdot N_{D,w} \cdot K_Q \cdot K_{P/W} \cdot K_{Vol}$$

Every factor is resolved automatically from your libraries and the active protocol, or entered on the **Factors** tab. Divisions are guarded, so a missing input *withholds* the dose rather than producing a wrong one.

Where the dose is reported — your choice, per machine

Set each machine's **dose-reporting point** alongside its SSD/SAD setup: report at **d_{max} / z_{ref}** — scaling the reference-depth dose by `PDD` (SSD) or `TMR` (SAD) — or **at the reference (calibration) depth** itself. The chosen depth travels with every saved record.

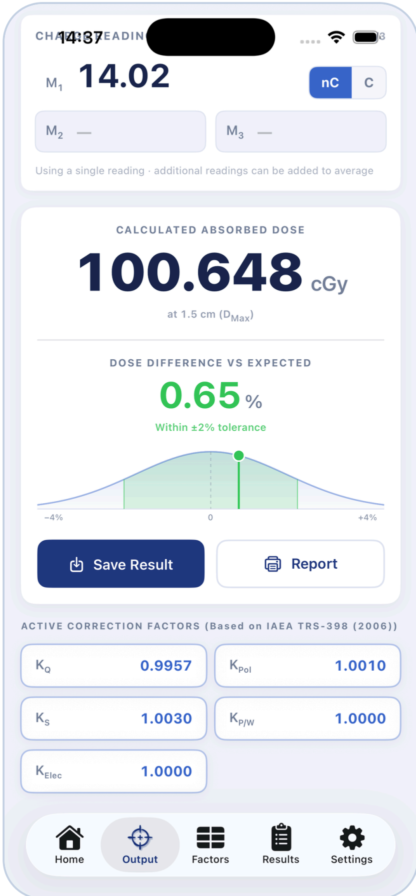
How the deviation reads

Output is compared as a rate — measured `cGy/MU` against the beam's calibration condition, so a per-MU calibration and a 100-MU calibration compare correctly. The % difference sits on a bell-curve gauge with a tolerance band:

- ≤ 2 % — within tolerance
- ≤ 3 % — investigate
- > 3 % — out of tolerance

Small & special fields

For machines with no 10×10 reference field (CyberKnife, TomoTherapy), the app routes to **TRS-483** small-field dosimetry ($k_{Qmsr,fref}$). Off-reference geometries withhold the dose until the reference field is restored. Please use these special-field machines with care — they are not yet fully validated, and your feedback is welcome and will be applied where needed.



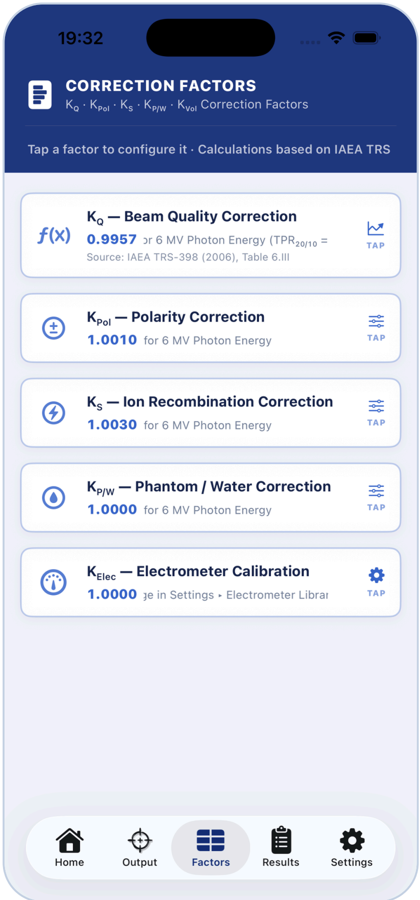
The Output tab — absorbed dose, deviation vs expected, and the active correction factors.

04 The correction factors

Nine quantities can enter a calculation. The app resolves each where it can, and the **Factors** tab is where you measure and *lock* the rest — stored per chamber and energy, so the same probe carries its own factors across every beam. Symbols follow each protocol's own notation (IAEA `K` / AAPM TG-51 `P`).

SYMBOL	CORRECTS FOR	WHERE IT COMES FROM
K_{TP} / P_{TP}	**Air density**	Live from temperature & pressure on Output
K_{Pol} / P_{Pol}	**Polarity effect**	Factors — opposite-polarity readings, per chamber+energy
K_{S} / P_{Ion}	**Ion recombination**	Factors — two-voltage method
K_{Elec} / P_{Elec}	**Electrometer calibration**	Electrometer library (default 1.000)
$K_{P/W}$	**Waterproof sleeve / phantom**	Factors — per chamber+energy
K_{Vol}	**Volume averaging (FFF beams)**	Automatic — needs cavity length (TRS-398 Rev.1)
$N_{D,w}$	**Absolute dose-to-water calibration**	Detector library (^{60}Co or cross-cal certificate)
K_{Q}	**Beam quality vs ^{60}Co**	Protocol tables, or a custom / overridden value
P_{Gr}	**Gradient (electron)**	Factors — AAPM TG-51 electron only

Factors are **locked to apply**: a measured factor stays out of the dose calculation until you lock it, so a half-entered value never silently changes a result. The Factors tab is a set of sheets on iPhone and a single-open accordion on iPad and Mac.



The Factors tab — every correction factor with its source; tap one to configure it.

05 Module map — where data flows

The Reference Dosimetry feature is four tabs. Data flows one way: configure once, calculate, save, report.

REFERENCE DOSIMETRY · TABS & DATA FLOW

SETTINGS

Libraries

Facilities, machines, detectors, electrometers, beams, calibrations.

↓

OUTPUT

Calculate

Enter the reading & live conditions; get dose and output deviation live.

↓

RESULTS

Save & trend

Auditable records, date filters, a scrolling trend chart.

↓

REPORT · BACKUP

Share & preserve

A4 PDF report; export/import everything as one file.

Factors sits alongside Output, feeding the corrections of section 04 into the calculation. When a chamber or beam changes on Output, the charge readings clear automatically so a stale number can't carry over.

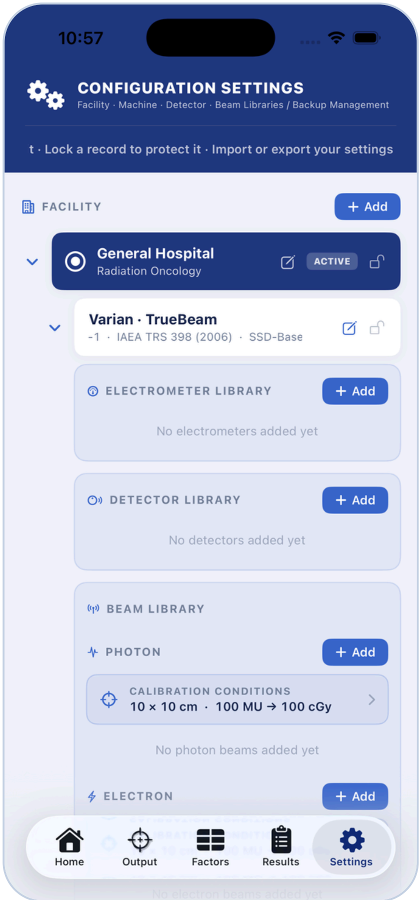
A measurement, start to finish

1. **Configure once**
   In Settings, add your facility, machine (protocol, setup, dose-reporting point), chambers with their $N_{D,w}$ certificates, and beam energies with their calibration conditions.

2. **Read & enter**
   On Output, pick the chamber and beam, then enter temperature and pressure (from live telemetry) and up to three charge readings.

3. **Lock & read**
   Absorbed dose and % deviation appear live on the gauge; lock any outstanding factors on the Factors tab.

4. **Save & report**
   Save the record, then filter, chart, and export a PDF — or back the whole database up to one file.



Facility → machine → detector / electrometer / beam libraries — the data hierarchy.

06 Protocols & beam quality

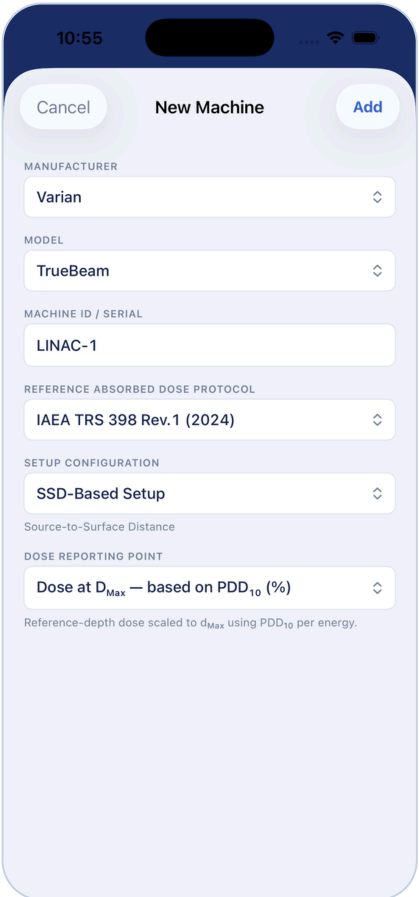
Each machine is assigned a code of practice; the engine routes the beam-quality factor and reference conditions accordingly. Three are live, and each expects a different beam-quality index:

PROTOCOL	PHOTON QUALITY	ELECTRON	NOTES
IAEA TRS-398 Rev.1	$TPR_{20,10}$	R_{50}	Default · auto K_{Vol} for FFF
IAEA TRS-398 (2006)	$TPR_{20,10}$	R_{50}	Tabulated K_Q · 5 or 10 cm depth
AAPM TG-51	$\%dd(10)_x$	R_{50}	Muir & Rogers 2014 electron K_Q

- **Switch safely.** Change a machine's protocol and any beam missing the new index (e.g. a $\%dd(10)_x$ where only $TPR_{20,10}$ exists) is flagged red with a clear message, before it can produce a wrong K_Q.
- **Custom K_Q.** Where a protocol's own tables can't answer — an unlisted chamber, or a beam quality outside the fitted range — supply your own value, stored per protocol and energy, never mixed between them.
- **Override K_Q.** When the tables *do* answer but you have an independently derived value you'd rather use, override the auto value from the K_Q card. Overriding is discouraged and clearly flagged — the

dose is tagged **"Overridden"** so its provenance is never hidden. Custom (a gap-filler) and Override (a replacement) stay distinct.

- **Coming soon.** DIN 6800-2 and NCS Report 18 appear in the picker, disabled for now.

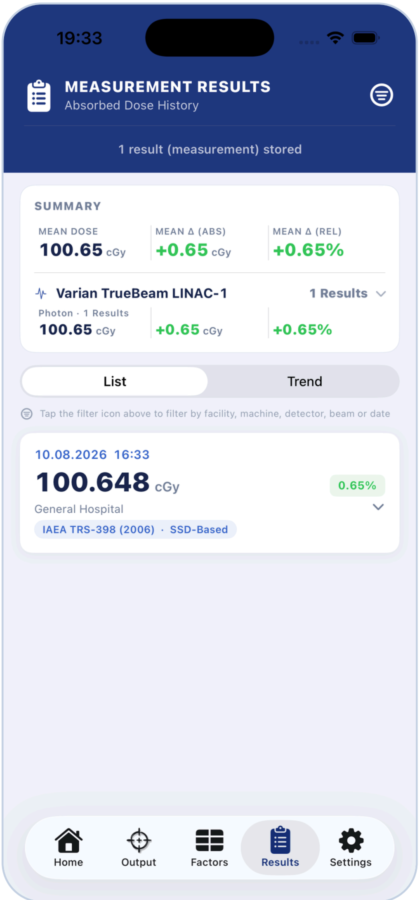


Per machine — reference dose protocol, setup configuration, and dose-reporting point.

07 Managing the results

A calculation isn't finished until it's recorded. The management side is as much the point as the math — every saved result keeps the full factor snapshot that produced it, down to the charge reading.

- **Records** — every saved calculation with its complete factor set, the mean charge reading M, $N_{D,w}$, and conditions; filterable by date (presets or a custom calendar range) and machine.
- **Trend chart** — a scrolling day-by-day view of output deviation with per-day labels, so drift is visible at a glance.
- **Report** — an A4 landscape PDF (trend chart + summary table, incl. M and $N_{D,w}$) built from the records you select.
- **Backup** — export everything (facilities, machines, detectors with certificates, beams with factors, saved results) to one file, and import it on another device; the format tolerates version changes.



The Results tab — saved measurements with summary, filterable list, and trend chart.

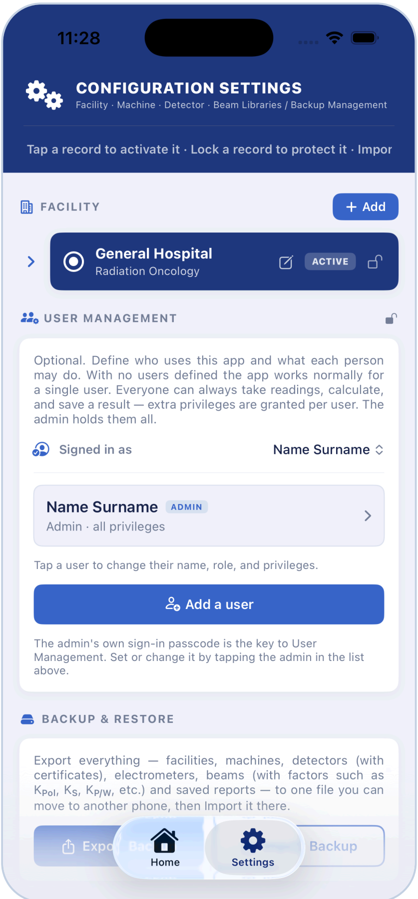
08 Users & sign-in `OPTIONAL`

For a **shared device**, RadiX Dose can record who performed each measurement. It is entirely optional — with no users defined the app behaves exactly as before. There is exactly **one admin** (the owner), who manages the roster in *Settings → User Management*; everyone else is an operator.

- **One admin, per app** — the first user created is the admin and holds every privilege. The app enforces a single admin: the Admin toggle is unavailable once one exists.
- **Privileges are additive** — taking readings, calculating, and *saving a result* are always allowed for everyone. Only extra powers are granted per person: editing the library, deleting results, and editing correction factors. A blocked action can be run on the spot by entering the admin passcode.
- **Passcodes & sign-in** — each user may have a numeric passcode. Once any user exists, every launch (and after the screen locks) shows a “Who is using RadiX Dose?” gate; the chosen operator's name is stamped as `Performed by` on each saved result — shown in the Results detail and the report PDF.
- **Admin recovery** — setting the admin passcode reveals a one-time recovery code (shown once). It is the only way to reset a forgotten admin passcode. A normal user who forgets theirs asks the admin to reset it.
- **Local only** — names and passcodes stay on the device (passcodes as one-way salted hashes) and are never transmitted. The `Performed by` stamp is a record-keeping aid, not a validated audit trail.

Admin lock

User Management, and any privileged action, unlock only for the admin (or with the admin passcode). Switching operators, locking the screen, or backgrounding the app re-locks it automatically.



User Management — the roster and the signed-in operator; tap a user to set privileges.

09 TPS Commissioning IN-DEVELOPMENT

⚠ Preview feature — not for clinical release

This feature is actively being built. Its workflow, tolerances, and acceptance logic are **not yet validated for clinical commissioning decisions**. Use it to explore the concept — not to sign off a treatment planning system. Reference Dosimetry is the released tool, verified against several known cases; this is not.

The idea: reuse the very same dose engine to turn a chamber reading at a point into a measured dose D_{meas}, compare it to the TPS-predicted dose, and evaluate the deviation against a tolerance — the standard point-dose validation of IAEA TRS-430, using the reference dose from IAEA TRS-398 or AAPM TG-51.

COMMISSIONING FLOW — PREVIEW

ORGANISE

Test Group

A TPS + algorithm + version, per machine.

↓

DEFINE

Test

A test type, energy, field size & setup, with measurement points.

↓

MEASURE

Point → D_{meas}

Chamber readings run through the reference dose engine.

↓

EVALUATE

vs TPS · pass/fail

Deviation against tolerance; confidence limit across a group.

Four test types

- **Central-axis (CAX) point dose** — the on-axis test.
- **Off-axis point dose (in field)** — a test inside the field edges.
- **Out-of-field point dose** — a test outside the beam.
- **Output factors** — the field-size output check.

How a point is judged

Each point takes up to three chamber readings (mean → `D_meas`) and the TPS-predicted dose. The deviation is plain for every point, evaluated against a tolerance you pick (1, 1.5, 2, 3 or 4 %), and a confidence limit summarises a whole group:

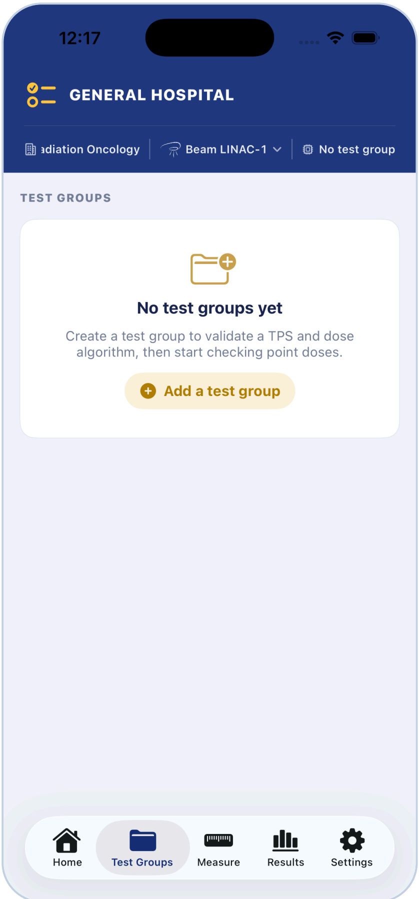
deviation

$$\delta = 100 \cdot (D_{TPS} - D_{meas}) \,/\, D_{meas}$$

confidence limit

$$CL = |\text{mean } \delta| + 1.5 \cdot SD$$

Because D_{meas} comes from the same verified chain as Reference Dosimetry, the measured side is sound — it is the surrounding commissioning workflow and acceptance logic that is still in development.



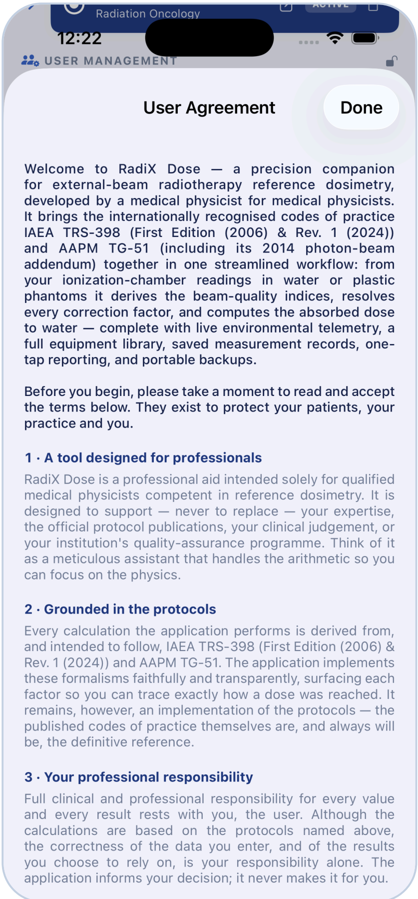
The TPS Commissioning workspace — Test Groups, Measure, and Results tabs (preview).

10 Your responsibility

RadiX Dose is a calculation aid. It launches behind a User Agreement you accept before clinical use, and the calculating physicist remains responsible for verifying every result against an independent method. The app warns — expired calibrations, missing inputs, off-reference geometry, mismatched beam quality, an overridden K_Q — but never overrides your judgement, and never blocks a calculation you choose to make.

In one line

Reference Dosimetry is ready for clinical use — to check and track output — with normal physics verification. **TPS Commissioning** is a preview of where RadiX Dose is heading — explore it, don't rely on it yet.



The User Agreement — accepted before clinical use; the physicist stays responsible.